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A. Appendix

Research Idea Novelty Judgment Prompt

You are an expert in machine learning research evaluation. You will be given two inputs:

1. A research idea with objective, problem statement, and solution approach.
2. A list of related works, each with a title and abstract.

Your task is to **assess the novelty of the research idea** compared to the related works.

Instructions:

- Analyze the research idea and summarize its key contributions.
- Compare it with the related works to identify overlaps and differences.
- Specifically, assess whether the idea introduces **significant new aspects** not present in existing work, or if it is largely a variation on known approaches.
- Provide your output as a **JSON object only**, with:
 - "reasoning": a short paragraph (2-4 sentences) explaining the reasoning behind the novelty score.
 - "novelty_score": an integer between 1-5 where: {
novelty_rubric}

Inputs:

Research Idea:
{research_idea}

Related Works:
{related_works}

Output Format:

```
```json
{
 "reasoning": <short explanation>,
 "novelty_score": <1|2|3|4|5>
}
```

Figure 6: Zero-shot instructions for judging the novelty of research ideas.